

Elliot D'Alessandro

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SUMMARY

UC Berkeley first-year (Data Science and Physics) pursuing research in theoretical CS, quantum computing, and physics, with prior research experience in query complexity and a background in systems programming and technical leadership. Experience leading a 7-person team across 6 games over three years, including multiple award-winning titles at Oregon's state competition and top 2% placement at the world's largest game jam.

RESEARCH

Theoretical Computer Science Research Assistant

Portland State University

Research Apprentice — Dr. Shravas Rao

2024

- Researched algorithmic complexity and quantum computing, focusing on algorithm power via query complexity.
- Worked with spectral sensitivity, relating structural graph properties to computational limits.
- Presented research findings to a cohort of 40 program interns in a formal end-of-program presentation.

EXPERIENCE

Game Developer and Systems Programmer

Portland, OR

Fractal Development

2022–2026

- Co-founded and grew a student development team from 3 to 7 members, directing all Unity and C# systems programming and audio composition.
- Designed and developed 3 games over three years for Oregon's statewide competition (50+ teams annually), winning 4 awards total: Best in Show, Player's Choice, Game Design, and Programming, becoming the first team to win both Best in Show and Player's Choice in the same year.
- Built 3 games for the world's largest game jam—the GMTK Game Jam, improving placement each year: top 7%, top 4%, then top 2% of 10,000+ entries.
- Built mobile app for high school information integration with Figma and Flutter, using HTTP request analysis to find API endpoints.

Technical Camp Counselor

Hillsboro, OR

Code Ninjas

Summer 2026

- Teach coding and STEM curriculum to campers ages 7–14 across Scratch, JavaScript, Roblox/Lua, and Minecraft modding, guiding them through Code Ninjas' belt-based skill progression.
- Guide students through hands-on projects, managing classroom flow and engagement across multiple skill levels simultaneously.
- Collaborate with fellow instructors and center staff to deliver a consistent, high-energy camp experience.

Course Development and Teaching Assistant

Portland, OR

Ida B. Wells High School

2025–2026

- Support instruction for a class of approximately 30 students; contribute to curriculum design, assignments, and lesson planning.
- Work one-on-one with students to explain concepts and debug code, adapting explanations until each concept clicks.

Math Tutor

Portland, OR

Multnomah County Library

2025–2026

- Tutor 6 students across 8 weeks in math spanning 4th–12th grade, adjusting approach to each learner's age and level.
- Foster an encouraging, low-pressure environment where students feel comfortable making mistakes and asking questions.

SKILLS

Programming Languages: Python, Java, JavaScript, C#, C++, HTML/CSS

Tools: Git, VS Code, Unity, LaTeX, Web Development, Confluence, Miro, Blender, FL Studio, DaVinci Resolve

General: Systems development, technical communication, project management, curriculum development

EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Data Science and B.S. Physics

2026–2030

Ida B. Wells High School

Portland, OR

High school diploma

2022–2026

Valedictorian, Rank 1/353, GPA: 4.388

ACTIVITIES, VOLUNTEERING & ATHLETICS

Athletics: Varsity Track & Field (4x Varsity Letter), Cross Country, Soccer

Volunteering: Library volunteer; track meet support (middle school & NCAA); trail restoration

Music: OMTA Syllabus Level 10; performed at Oregon Federation Junior Festival, 3 gold cup awards